Scoped O.1 on the Cone, v3: Regularisation Invariance at Generic Opening Angle, the Inner-Resonance Obstruction, the Contact Tadpoles D05a and D12, Resonant Trilinear Cohomology, and One-Sided Rigorous Rank Bounds

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Abstract

We continue the transport of the scoped closure O.1 of the Enter companion programme (XVI-Q record Q11: one loop, order κ^2, bulk two-point sector, $m = 0$, frozen de Donder gauge, XVI-P prescription) to the flat cone $C_\alpha \times \mathbb{R}^2$ on the curvature-regular (CR) branch. This version is self-contained and supersedes v2; all v2 results are kept (two corrections, §3). New in v3: (1) a proof sketch of the regularisation hypothesis ($\mathrm{H}_{\mathrm{reg}}$) for the bubble graphs at every *inner-non-resonant* α, resting on a computed classification of the one-point (inner) apex degrees of the bubble integrands, together with the precise obstruction at the *inner-resonant* set, which is exactly $\{1/2, 2/3, 4/5\}$ on CR and $\{1/2\}$ on F (a new resonance, 4/5, absent from the two-point inventory); (2) for the bubbles D01, D03, D04, D06, D11, D13: a reduction of the apex pole to a single Mellin residue, with D04 and D11 shown inner-clean at every α; the residues themselves are *not* evaluated, so the 6 exact and 2 $\phi\phi$ values remain open; (3) the contact tadpoles D05a ($\phi\phi$, both branches) and D12 (hh from the $hh\phi\phi$ vertex, scalar loop, branch-independent) in closed form in α, and D02 (hh from the quartic vertex, both branches, 48 exact coefficients); (4) evanescent apex coefficients: still open (they need the bubbles); (5) the three-field ghost-number-one apex cohomology at the resonant $\alpha \in \{1/3, 1/2, 2/3\}$ with *all* lattice jets: $H^1_{(3)} = 0$ at each, two primes; (6) the modular rank certificates are upgraded to *rigorous one-sided* bounds, which already make every "$H^1 = 0$" a rigorous upper-bound statement once a single spanning hypothesis holds; a third prime agrees everywhere. Fully exact (fraction-free) certificates were not produced. The enter-companion registry was re-fetched: 249 sheets, all sha256-identical to the previous intake, so no new input was available. **O.1 on the cone remains NOT_CLOSED.**

1 Background, scope and conventions

Enter background. The Enter companion programme (repository `cheungngo/enter-companion`) is a sequence of machine-checked landmarks about perturbative gravity coupled to a scalar on fixed backgrounds. XVI-P fixes the prescription (DRED with quasi-d-dimensional loop momenta and quasi-4-dimensional vertex indices, $\overline{\mathrm{MS}}$, the p22 evanescent-counterterm contract). XVI-Q certifies the scoped closure O.1 (Q03/Q11), and XVI-R

imports Q11. All three trees are read-only inputs; we used copies in `/tmp/o1cone_ro`, and the tree hashes of the originals agree before and after this work (§12). The cone and its CR extension (the I-B-G-S v2 spin-2 Green operators) are *inputs*, not consequences. A separation is not a bridge; cited sheets are not promoted to theorems.

Scope. *O.1 on the cone* is the Q11 statement with Minkowski space replaced by $M_\alpha = \{-dt^2 + dr^2 + \alpha^2 r^2 d\varphi^2 + dz^2\}$, $t := 1/\alpha > 1$, apex $W = \{r = 0\}$. Sector, order, prescription and gauge are unchanged; spin-2 fields obey CR; Friedrichs (F) is used only for comparison. `bulk_symmetry_restored` = `false`.

Kernels and lattice (from v1/v2). The helicity-S kernels are $G_b = S_b(u,\psi)/(8\pi^2\alpha\, rr' \sinh\eta)$ with $S_b = f_b(u - i\psi) + f_{1-b}(u + i\psi)$, $f_b(\zeta) = e^{-b\zeta}/(1 - e^{-\zeta})$, labels $b(S) = S\alpha$ $(S \geq 0)$, $1 + S\alpha$ $(S < 0)$ on CR and $\mathrm{frac}(S\alpha)$ on F; Synge data A, B, C are exact (Bernoulli form). Jets near W are (C, n, m) with rotation charge n and radial power m: $\nu = |nt + \sigma|$, $m = \nu + 2j$, except in the CR-flipped channels $(\sigma, n) = (\pm 2, \mp 1)$ for $t < 2$, where $m = -\nu + 2j = t - 2 + 2j$.

2 Step 0: registry re-intake

The registry (`/browse` and `/api/manuscript/<id>`) was re-fetched read-only into `/workspace/XVI-R-companion-intake/new_oct18c/`: 249 ids, the same id set as the previous intake `new_oct18b`, all 249 fetched with return code 0. Every sheet is sha256-identical to `new_oct18b/all`: **0 new, 0 changed** (`index.json`, `SHA256SUMS`, `SHA256SUMS_all`). The GitHub API for the repository returns 404, so the site is the only source. Consequently nothing new could help. The only relevant existing inputs remain I-B-VP-1 (v3: existence/profile-independence of the extension U_L of r^{-2}, simultaneous Ward-compatible extension, complete ambiguity by δ_W contact terms), I-B-VP-1-Ex1 (existence-and-residue lemma) and I-B-VP-6 (extension ledger). These concern *scalar one-point* apex extensions. We use them only as cited motivation for the one-point (inner) extension step in §5; the step itself is proved there from Hörmander's theorem, not from the sheets. They say nothing about two-point bubble products, so they do not supply the values in item (2).

3 Corrections to v2

1. *F-channel wording in the structure theorem.* v2 Thm. 3.2(i) said "under F the same channels with $t < 2$ carry $A = -1$". Wrong as written: under F these channels have $m = \nu = 2 - t(+2j)$, i.e. the coefficient of t is -1 (not $|n| = +1$), and the constant is $+2$. The decomposition $m = |n|t + A$ with integer A holds on CR only; on F the flipped-sign channels have $m = -|n|t + 2 + 2j$. No count or conclusion of v2 depended on the misstatement (the F rows were diagnostic, and the resonance scan in `code/resonances.py` used the correct ν).

2. *Resonances are sector-dependent.* v2 called $\{1/3, 1/2, 2/3\}$ "the resonant set". That is correct for two-point apex *candidates* ($T \leq 5$). It is not the set relevant for the bubble integrands: the one-point (inner) degrees of the bubbles have integer non-integrable values on CR exactly at $\{1/2, 2/3, 4/5\}$ (Theorem 5.1). $\alpha = 4/5$ is new and $1/3$ is not inner-resonant. The v2 claim that the counts in the two-point inventory are α-independent off $\{1/3, 1/2, 2/3\}$ stands.

3. *Ranks.* v2 described the modular ranks as Schwartz–Zippel certificates with a false-drop probability. v3 (§10) shows that half of every such statement is in fact rigorous (one-sided), and isolates the one probabilistic hypothesis.

4 Results carried over from v1/v2 (re-verified where re-run)

- Regular-region locality (v1): on $C_c^\infty(M^2_{\mathrm{reg}})$ the one-loop pole equals the flat XVI-Q one (21 directions, flat coefficients), both branches.
- FM/Kabat coefficient matches (v1), exact.
- Structure theorem and two-point inventory (v2; corrected wording above): on CR, $hh/\bar{c}c/Kc/\phi\phi = 18/8/6/2$, $N = 34$, $\dim Z = 8$, $\dim B = 6$, $H^0_{(2)} = 2$, gh1 closed=exact= 26, $H^1_{(2)} = 0$ off $\{1/3, 1/2, 2/3\}$; $(35, 8, 6, 2; 27/27)$ at $1/3$ and $2/3$, $(42, 9, 7, 2; 33/33)$ at $1/2$. All re-run in v3 at a third prime, $p_3 = 10^9 + 21$ ($\equiv 1 \bmod 4$), identical (`logs/v2apex/apex_*_p2.json`).
- $E = 1$ row (v2 Thm. 5.1): D14a,b,c apex poles; scalar loop = $\frac{\kappa}{16\pi^2} a_2^\Sigma$ (Fursaev), graviton+ghost closed on CR, not closed on F; $\varsigma = -1$ applied.
- Generic three-field cohomology (v2): $H^1_{(3)} = 0$, 83/83 of 194; re-run at p_3, identical (`logs/v2_tri_7071_10000_p2.json`).

5 Item (1): the regularisation hypothesis ($\mathrm{H}_{\mathrm{reg}}$) for the bubbles

Inner degrees. Consider a bubble graph with cubic vertices at x and x'. Let $x \to W$ with x' fixed away from W. Near W each propagator leg at x (two internal, one external) is a sum of jets $r^m e^{in\varphi}$, and at most $k \le 2$ transverse derivatives act at the vertex. The integrand is therefore a finite sum of terms $r^s e^{iN\varphi} \times$(smooth in the longitudinal variables) with

$$s = m_1 + m_2 + m_3 - k, \qquad N = n_1 + n_2 + n_3,$$

and only $N = 0$ survives the angular integral. The term is locally integrable in the 2-dimensional transverse plane iff $s > -2$. Terms built only from $n = 0$ jets are products of derivatives of smooth functions ($r^{|\sigma|}e^{i\sigma\varphi}$ in the helicity frame are polynomials), hence smooth; they cause no apex singularity.

Theorem 5.1 (Inner-degree classification)**.** *Computed exhaustively in* `code/inner_degrees.py` *and* `code/inner_degrees_q24.py` *for the vertices of D01* $(h, h|h)$*, D03* $(c, c|h)$*, D04* $(h, \phi|\phi)$ *and* $(\phi, h|\phi)$*, D11* $(\phi, \phi|h)$ *and D13* $(c, h|K)$*,* $(c, h|c)$ *(D06 is the D01+D03 channel), with* $|n| \le 5$*,* $j \le 1$*, on the grid* $\{k/97\} \cup \{p/q : q \le 24\}$ *(276 values), both branches:*

(i) On CR, a non-integrable inner degree ($s \le -2$*,* $N = 0$*, some* $n_i \ne 0$*) that is an* integer *occurs exactly at* $\alpha \in \{1/2, 2/3, 4/5\}$*. The affected graphs are D01 (6 terms at* $1/2$*, 42 at* $2/3$*), D03 (4 at* $2/3$*), D13 (*$2 + 0$ *at* $1/2$*,* $16 + 4$ *at* $2/3$*), and at* $4/5$ *D01, D03 and D13 (example:* $n = (1, 1, -2)$*,* $s = 4t - 6 - k = -2$ *at* $k = 1$*).*

(ii) On F this happens only at $\alpha = 1/2$ *(where F and CR coincide).*

(iii) D04 and D11 have no non-integrable inner degree with $n \ne 0$ *at any* α*; they are* inner-clean*.*

(iv) At every other α all non-integrable inner degrees are non-integer. At $\alpha = 7071/10000$ the minima are $s_{\min} = -3.1715$ (D01, D13a) and -2.1715 (D03, D13b).

The grid is complete. For $\alpha < 1/2$ there are no flips, all $m \geq 0$, and an integer $s \leq -2$ needs $k = 2$ and three $m = 0$ jets, one with $n \neq 0$, i.e. $\nu = |nt+\sigma| = 0$, impossible for $t > 2$. At $\alpha = 1/2$ ($t = 2$), $\nu = 0$ at $(\sigma, n) = (\pm 2, \mp 1)$ gives the non-smooth $r^0 e^{\mp i\varphi}$. For $\alpha \in (1/2, 1)$, each flip has $n = \mp 1$ and $m \geq t - 2 > -1$; three flips cannot satisfy $N = 0$, so at most two occur and contribute > -2. Since $k \leq 2$, $s \leq -2$ forces every non-flipped jet to have $m < 2$, and $m \geq |n|t - 2$ gives $|n| < 4/t < 4$. Hence $s = At + B$ with $|A| \leq 9$, and $s \in \mathbb{Z}$ forces $\alpha = |A|/K$ with $|A| < K < 2|A| \leq 18$. All such values lie in the grid.

The degree count is power counting: vertex tensor contractions were not evaluated. They can only remove terms, so the lists in (i) are upper bounds and (iii), (iv) are unaffected.

Proposition 5.2 (($\mathrm{H}_{\mathrm{reg}}$) at inner-non-resonant α, CR; modulo (J))**.** *Let $\alpha \notin \{1/2, 2/3, 4/5\}$ on CR. Define the bubble regularisation as: longitudinal dimensional continuation (XVI-P DRED, $d - 2$ longitudinal dimensions, transverse plane exactly 2-dimensional), composed with, at each vertex approaching W alone, the* unique homogeneous extension *of every non-integrable inner term $r^s e^{iN\varphi}$ (unique because $s \notin \{-2, -3, \dots\}$; Hörmander, Thm. 3.2.3). Assume*

(J) the joint region $x, x' \to W$ is regulated by d, i.e. its joint-scaling Mellin exponent depends on d non-degenerately.

Then the regularised one-loop κ^2 two-point functional satisfies the linearised Slavnov–Taylor identity up to terms $(d-4)\times$(regularised), so its pole part is σ_0-closed. Hence (v2 Thm. 6.1) the 26 Ward-constrained pole coefficients vanish (27 at 1/3).

Proof sketch. (a) *Domain.* On CR the BRST variations map lattice jets into lattice jets ($K_2 c$ has no foreign atom in any sample; v1/v2). (b) *Commutation.* The unique homogeneous extension E_s of degree s commutes with ∂_i ($E_{s-1}\partial = \partial E_s$, both are homogeneous extensions of ∂f and $s - 1$ is again non-exceptional) and with multiplication by x_i (degree $s + 1$, non-exceptional). The Ward identities of the propagators hold as distributions including W (v1). Hence every algebraic step of the flat ST derivation goes through for the extended integrands. (c) *Longitudinal DR* commutes with longitudinal derivatives and translations. The only breaking is the mismatch between quasi-d momenta and quasi-4 indices, which is evanescent, $(d-4)\times$ an object with at most a simple pole, hence finite at one loop. (d) By (J) the joint region produces a pole in d only, and it inherits (a)–(c). Finite apex breakings are WZ-closed and exact by $H^1_{(2)} = 0$, so they are absorbed by finite inventory counterterms. □

Status and obstruction. (J) is argued, not proved. A proof needs the joint Mellin representation of §6 and a check that its exponent is not d-independent for some channel. At $\alpha \in \{1/2, 2/3, 4/5\}$ the obstruction is precise. The listed integer terms with $N = 0$ have non-vanishing angular moment ($\int d\varphi \neq 0$ at $s = -2$), so there is no homogeneous extension: they are genuine transverse logarithms that longitudinal DR does not regulate. They need either an extra regulator (e.g. analytic $r^{2\epsilon}$, whose BRST behaviour is not established) or cubic apex counterterms on W. If the breaking induced by such a regulator is local on W and WZ-consistent, it is removable when the three-field cohomology vanishes at that α. We computed $H^1_{(3)} = 0$ at 1/2 and 2/3 (§9), and at 4/5 (all-jet run, candidate counts generic; §9). That gives a conditional route, not a proof. On F, ($\mathrm{H}_{\mathrm{reg}}$) fails already for the $E = 1$ row (v2), so no F statement is made. **Item (1) outcome: partially earned**: proved modulo (J) and standard lemmas at inner-non-resonant α; obstructed (precisely located) at $\{1/2, 2/3, 4/5\}$.

6 Item (2): apex poles of the bubbles

Reduction. Let $B(x,x')$ be a bubble integrand after the inner extensions of §5. Write $x=(\rho\omega,y)$, $x'=(\rho\omega',y')$ with (ω,ω') on the unit scaling sphere Σ of the transverse pair $(x_\perp,x'_\perp)\in\mathbb{R}^2\times\mathbb{R}^2$ (dimension 3). Then the apex pole of the bubble is

$$\mathrm{Pole}_{d=4}\,\Gamma_W=\frac{1}{4-d}\,\mathrm{Res}_{z=z_*(d)}\int_\Sigma d\omega\,d\omega'\;\mathcal{M}[B](z;\omega,\omega';p_\parallel),$$

where $\mathcal{M}$ is the Mellin transform in ρ, taken after the bulk Hadamard subtraction at $x=x'$ (which reproduces the flat 21-direction pole of Theorem v1) and after the inner extensions. Projected onto the 34-dimensional basis (equivalently, by v2 Thm. 6.1 and Prop. 5.2, onto the 8 closed directions), it gives the 6 exact and the 2 $\phi\phi$ values. For D04 one can do better. Split the graviton propagator as $\Pi\,G_0+\sum_S\Pi_S(G_{b(S)}-G_0)$. The first part obeys the flat DRED identity $V\Pi V'=2p^2(l-p)^2$ and reduces to a tadpole (route of §7). The second part has a bulk-smooth kernel $G_{b(S)}-G_0$, so its pole is purely apex, with no Hadamard subtraction. D04 and D11 need no inner extension at any α (Thm. 5.1(iii)).

Obstruction. The residue is a 3-dimensional singular integral over Σ of products of the exact Bernoulli kernels (and of their first and second derivatives), with an extra longitudinal loop integral in $d-2$ dimensions. We did not evaluate it, for D04 alone or D06 alone or any other bubble. The Synge (coincidence) data used for tadpoles do not suffice, because G_b-G_0 is not smooth at $r=r'=0$. The I-B-VP-1 technology handles one-point extensions (U_L of r^{-2}, ambiguity δ_W) and is used in the inner step. It does not provide the joint two-point residue. **Item (2) outcome: open**, with the reduction above. The 26 Ward-constrained combinations vanish only modulo (J) (§5). The 6 exact and 2 $\phi\phi$ values are *not* computed (missing $\neq$ zero).

7 Item (3): the contact tadpoles D12, D05a, D02

Method. This is the coincidence route of v2 §5 with the corrected sign $\Gamma=-\text{prefactor}\cdot[\dots]$ ($\varsigma=-1$), implemented in `code/tadpole2.py` and, with term-wise extraction of the λ^{-2} coefficient, in `code/tadpole2_fast.py`. The fast version reproduces all 18 D12 coefficients of the slow one exactly. The external fields are the most general rotation-equivariant, reflection-even $n=0$ jets up to $m=2$, each times $e^{\pm i(p_0t+p_3z)}$:

- $h_{00}=a_0+a_1r^2$, $h_{03}=b_0+b_1r^2$, $h_{33}=c_0+c_1r^2$, $h_{0i}=d\,x_i$, $h_{3i}=e\,x_i$, $h_{ij}=\delta_{ij}(f_0+f_1r^2)+g(x_ix_j-\frac12\delta_{ij}r^2)$;
- $\phi=s_0+s_1r^2$.

We evaluate on two rays, $(1,0)$ and $(3/5,4/5)$. θ-independence was checked in every case. The output is the coefficient of $(4-d)^{-1}$ per unit area of W, as an exact bilinear in the two external fields (subscripts $1,2$).

Theorem 7.1 (D12 on the cone)**.** *D12 is the $hh\phi\phi$ vertex closed by a scalar loop, prefactor $-\kappa^2/2$ (XVI-P). Its loop label is $b=0$, so it is the same on CR and F. Set*

$$A_\alpha=\frac{\kappa^2(\alpha^2-1)(4\alpha^2-1)}{1440\pi\alpha^3},\qquad C_\alpha=\frac{\kappa^2(\alpha^2-1)(11\alpha^2+1)}{1440\pi\alpha^3}.$$

Then, with $(1 \leftrightarrow 2)$ *denoting the symmetrisation,*

$$\begin{aligned}\operatorname{Pole}\Gamma_{D12} = A_\alpha\Big[-2\langle h_1^{(0)}, h_2^{(1)}\rangle_\parallel + \operatorname{tr}_\parallel h_1^{(0)}\operatorname{tr}_\parallel h_2^{(1)}\Big] - C_\alpha\, g_1 \operatorname{tr}_\parallel h_2^{(0)} + 2C_\alpha\, f_{0,1}\, g_2 \\ - 2C_\alpha\,(d_1 d_2 - e_1 e_2) \;+\; (1 \leftrightarrow 2).\end{aligned}$$

Here $h^{(0)}, h^{(1)}$ *are the* r^0 *and* r^2 *coefficients of the longitudinal block,* $\langle h, h'\rangle_\parallel = \eta^{ac}\eta^{bd}h_{ab}h'_{cd}$ *and* $\operatorname{tr}_\parallel h = \eta^{ab}h_{ab} = -h_{00} + h_{33}$*. In raw coefficients:* $a_0a'_1$*,* $a_0c'_1$*,* $c_0a'_1$*,* $c_0c'_1$ *all carry* $-A_\alpha$*;* $b_0b'_1$ *carries* $+4A_\alpha$*;* a_0g' *carries* $+C_\alpha$ *and* c_0g' *carries* $-C_\alpha$*;* f_0g' *carries* $+2C_\alpha$*;* dd' *carries* $-4C_\alpha$ *and* ee' *carries* $+4C_\alpha$ *(`logs/v3/tad2_D12_CR.json`). The result is symmetric under* $1 \leftrightarrow 2$ *and longitudinally* $O(1,1)$*-covariant (checked coefficient by coefficient). It is momentum-independent, so it lies entirely in the two-transverse-derivative (*r^2*-jet) apex directions. Every coefficient carries* $(\alpha^2 - 1)$ *and vanishes at* $\alpha = 1$*, as it must for a* $b = 0$ *loop.* f_1 *does not appear.*

Theorem 7.2 (D05a on the cone)**.** *D05a is the* $hh\phi\phi$ *vertex closed by a graviton loop, with external* $\phi\phi$*. Its prefactor is* $-\kappa^2/2$ *by analogy with D02/D12, because D05a is not in the XVI-P amplitude rules.* This normalisation is flagged. *Then*

$$\operatorname{Pole}\Gamma^{\mathrm{CR}}_{D05a} = -\frac{\kappa^2(2\alpha-1)}{4\pi}\, s_{0,1}s_{0,2}\,(p_0^2 - p_3^2), \qquad \operatorname{Pole}\Gamma^{\mathrm{F}}_{D05a} = -\frac{\kappa^2(\alpha-1)(2\alpha-1)}{4\pi\alpha}\, s_{0,1}s_{0,2}\,(p_0^2 - p_3^2).$$

Since $p_0^2 - p_3^2 = -\eta^{ab}p_a p_b$*, the result lies on the first* $H^0_{(2)}$ *direction* $\delta_W\phi\,\partial_\parallel^2\phi$*. The second direction,* $\delta_W\phi\,\partial_r^2\phi|_{m=2}$*, receives* 0 *from this graph (no* s_0s_1 *term). F vanishes at* $\alpha = 1$*. CR has the flat-limit remnant* $-\kappa^2/(4\pi)$*, which parallels the CR remnant of the* $E = 1$ *row (v2), and both branches vanish at* $\alpha = 1/2$*, where they coincide.*

D02. D02 is the quartic graviton vertex $V_{4,\mathrm{grav}}$ closed by a graviton loop, prefactor $-\kappa^2/2$ (XVI-P). It involves 19 384 vertex-monomial terms and was computed on both branches (`logs/v3/tad2fast_D02_CR.json`, `_F.json`, 158 s and 156 s). The pole has 48 nonzero bilinear coefficients on each branch, all exact rational functions of α times κ^2/π, listed in Table 1. Internal checks, all passed exactly:

- θ-independence;
- $1 \leftrightarrow 2$ symmetry (with $p \to -p$);
- longitudinal $O(1,1)$ structure, e.g. $-K_\alpha f_{0,1}f_{0,2}(p_0^2 - p_3^2)$ and $-K_\alpha f_0\,(a_0p_3^2 - 2b_0p_0p_3 + c_0p_0^2)$, with $K_\alpha = \kappa^2(2\alpha-1)(4\alpha-1)/(6\pi\alpha)$;
- F vanishes identically at $\alpha = 1$;
- CR = F at $\alpha = 1/2$.

CR has a nonzero $\alpha \to 1$ remnant (e.g. $a_0^{(1)}a_1^{(2)} \to -\kappa^2/4\pi$), like D14 and D05a.

Table 1: D02 apex pole: coefficient of each monomial (CR and F).

monomial	CR	F
$f_0^{(1)}f_0^{(2)}p_3^2$	$\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$a_0^{(1)}f_0^{(2)}p_3^2$	$-\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$-\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$

monomial	CR	F
$f_0^{(1)}a_0^{(2)}p_3^2$	$-\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$-\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$b_0^{(1)}f_0^{(2)}p_0p_3$	$\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{3\pi\alpha}$	$\frac{4\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$f_0^{(1)}b_0^{(2)}p_0p_3$	$\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{3\pi\alpha}$	$\frac{4\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$e^{(1)}f_0^{(2)}p_3$	$\frac{i\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$f_0^{(1)}e^{(2)}p_3$	$-\frac{i\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$a_0^{(1)}e^{(2)}p_3$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$b_0^{(1)}d^{(2)}p_3$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$d^{(1)}b_0^{(2)}p_3$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$e^{(1)}a_0^{(2)}p_3$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$f_0^{(1)}f_0^{(2)}p_0^2$	$-\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$-\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$c_0^{(1)}f_0^{(2)}p_0^2$	$-\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$-\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$f_0^{(1)}c_0^{(2)}p_0^2$	$-\frac{\kappa^2(2\alpha-1)(4\alpha-1)}{6\pi\alpha}$	$-\frac{2\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$d^{(1)}f_0^{(2)}p_0$	$-\frac{i\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$b_0^{(1)}e^{(2)}p_0$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$f_0^{(1)}d^{(2)}p_0$	$\frac{i\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$c_0^{(1)}d^{(2)}p_0$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$d^{(1)}c_0^{(2)}p_0$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$-\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$e^{(1)}b_0^{(2)}p_0$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$\frac{i\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$
$f_0^{(1)}g^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)\left(11\alpha^2+1\right)}{360\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)\left(11\alpha^2+1\right)}{360\pi\alpha^3}$
$c_0^{(1)}g^{(2)}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$
$a_0^{(1)}g^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$
$f_0^{(1)}f_1^{(2)}$	$\frac{\kappa^2(\alpha-1)\left(259\alpha^3-101\alpha^2-\alpha-1\right)}{180\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)\left(259\alpha^3-821\alpha^2+599\alpha-121\right)}{180\pi\alpha^3}$
$c_0^{(1)}f_1^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$
$a_0^{(1)}f_1^{(2)}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$
$g^{(1)}f_0^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)\left(11\alpha^2+1\right)}{360\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)\left(11\alpha^2+1\right)}{360\pi\alpha^3}$
$f_1^{(1)}f_0^{(2)}$	$\frac{\kappa^2(\alpha-1)\left(259\alpha^3-101\alpha^2-\alpha-1\right)}{180\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)\left(259\alpha^3-821\alpha^2+599\alpha-121\right)}{180\pi\alpha^3}$
$c_1^{(1)}f_0^{(2)}$	$-\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$-\frac{\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$a_1^{(1)}f_0^{(2)}$	$\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$e^{(1)}e^{(2)}$	$\frac{\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(\alpha^2-3\alpha+1\right)}{3\pi\alpha^3}$
$d^{(1)}d^{(2)}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)}{3\pi\alpha}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(\alpha^2-3\alpha+1\right)}{3\pi\alpha^3}$
$f_0^{(1)}c_1^{(2)}$	$-\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$-\frac{\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$
$c_0^{(1)}c_1^{(2)}$	$-\frac{\kappa^2\left(874\alpha^4-900\alpha^3+205\alpha^2+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)\left(874\alpha^3-386\alpha^2-121\alpha+59\right)}{720\pi\alpha^3}$
$a_0^{(1)}c_1^{(2)}$	$-\frac{\kappa^2(2\alpha-1)\left(178\alpha^3-\alpha^2+2\alpha+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(178\alpha^2-3\alpha-61\right)}{720\pi\alpha^3}$
$g^{(1)}c_0^{(2)}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$
$f_1^{(1)}c_0^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$
$c_1^{(1)}c_0^{(2)}$	$-\frac{\kappa^2\left(874\alpha^4-900\alpha^3+205\alpha^2+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)\left(874\alpha^3-386\alpha^2-121\alpha+59\right)}{720\pi\alpha^3}$
$a_1^{(1)}c_0^{(2)}$	$-\frac{\kappa^2(2\alpha-1)\left(178\alpha^3-\alpha^2+2\alpha+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(178\alpha^2-3\alpha-61\right)}{720\pi\alpha^3}$
$b_0^{(1)}b_1^{(2)}$	$\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)\left(41\alpha^3-19\alpha^2-8\alpha+4\right)}{12\pi\alpha^3}$
$b_1^{(1)}b_0^{(2)}$	$\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)\left(41\alpha^3-19\alpha^2-8\alpha+4\right)}{12\pi\alpha^3}$
$f_0^{(1)}a_1^{(2)}$	$\frac{\kappa^2\left(41\alpha^2-36\alpha+7\right)}{12\pi\alpha}$	$\frac{\kappa^2(\alpha-1)\left(41\alpha^2-43\alpha+12\right)}{12\pi\alpha^2}$

monomial	CR	F
$c_0^{(1)}a_1^{(2)}$	$-\frac{\kappa^2(2\alpha-1)\left(178\alpha^3-\alpha^2+2\alpha+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(178\alpha^2-3\alpha-61\right)}{720\pi\alpha^3}$
$a_0^{(1)}a_1^{(2)}$	$-\frac{\kappa^2\left(874\alpha^4-900\alpha^3+205\alpha^2+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)\left(874\alpha^3-386\alpha^2-121\alpha+59\right)}{720\pi\alpha^3}$
$g^{(1)}a_0^{(2)}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$	$\frac{\kappa^2(\alpha-1)(\alpha+1)(7\alpha-1)(7\alpha+1)}{720\pi\alpha^3}$
$f_1^{(1)}a_0^{(2)}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$	$-\frac{\kappa^2(\alpha-1)(\alpha+1)}{12\pi\alpha}$
$c_1^{(1)}a_0^{(2)}$	$-\frac{\kappa^2(2\alpha-1)\left(178\alpha^3-\alpha^2+2\alpha+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)(2\alpha-1)\left(178\alpha^2-3\alpha-61\right)}{720\pi\alpha^3}$
$a_1^{(1)}a_0^{(2)}$	$-\frac{\kappa^2\left(874\alpha^4-900\alpha^3+205\alpha^2+1\right)}{720\pi\alpha^3}$	$-\frac{\kappa^2(\alpha-1)\left(874\alpha^3-386\alpha^2-121\alpha+59\right)}{720\pi\alpha^3}$

What this gives for the eight closed directions. The $\phi\phi$ apex poles are D05a+D04. D05a is computed (above, normalisation flagged), and D04 is an open bubble (§6), so the two physical $\phi\phi$ values are *not* determined. The hh contact graphs D12 and D02 are computed in the explicit jet basis above. The bubbles D01, D03, D06, D11 and D13 are open. We did not map the explicit-basis bilinears to the engine's 18-dimensional hh coordinates. The two bases differ by total derivatives on W and by the $n = 0$, $m \leq 2$ truncation of the test fields, and this map is not implemented. So no explicit-basis cross-check of "$hh = 18$" is claimed. **Item (3) outcome**: D12 computed (exact; branch-independent); D05a computed (exact, both branches; prefactor flagged); D02 computed (exact, both branches).

8 Item (4): evanescent apex coefficients

DRED with quasi-4 indices and $(d-2)$-dimensional longitudinal momenta gives an evanescent twin ($\hat{p}^2$ for $p_\parallel^2$) for every $k = 2$ class (v2 §6: three in hh, one in $\bar{c}c$, one in $\phi\phi$). Their one-loop coefficients are finite, pole$\times(d-4)$, and need the full d-dependence of each graph's apex part. For the tadpoles that would require redoing the coincidence route with the d-dimensional trace kept symbolic, which is not done. For the bubbles it needs the open residues. **Item (4) outcome: open** (no coefficient computed).

9 Item (5): three-field cohomology at the resonant α

The v2 three-field engine used $n = 0$ jets only, which is valid off the resonant set. `code/tri_count_all.py` and `code/tri_run_all.py` use *all* lattice jets ($|n| \leq 4$) with $\sum n_i = 0$ and integer derivative budget. All other ingredients are unchanged: the minimal BV complex, the maps σ_0 of v2 §4, and the $\sigma_0^2 = 0$ check. As a control, at generic $\alpha = 7071/10000$ the all-jet raw counts equal the $n = 0$ counts (hhh 780, hhc 856, Kcc 114, $\phi\phi c$ 24, hcc 784), as Theorem v2-A3(iii) predicts.

α (CR)	raw hhh/hhc	gh0 dims $hhh, \phi\phi h, Kch, c^*cc, \phi^*\phi c$	gh1 dims $hhc, \phi\phi c, Kcc$ (total)	closed/exact	$H^1_{(3)}$
generic (v2; p_3 v3)	780/856	51, 12, 31, 4, 3	161, 10, 23 (194)	83/83	0
1/3	792/876	53, 12, 31, 4, 3	166, 10, 23 (199)	85/85	0
1/2	1134/1266	81, 13, 41, 4, 3	254, 11, 29 (294)	120/120	0
2/3	918/996	66, 12, 34, 4, 3	198, 10, 24 (232)	99/99	0
4/5 (inner-resonant)	780/856	51, 12, 31, 4, 3	161, 10, 23 (194)	83/83	0

Each row was run at p_1 and p_2 with identical results, and in each the images lie in the span and $\sigma_0^2 = 0$ holds (`logs/v3/tri_all_*.json`). At 4/5 the all-jet counts equal the generic ones, so 4/5 is not resonant for three-field candidates even though it is inner-resonant for the bubbles. **Item**

(5) outcome: computed: $H^1_{(3)} = 0$ on CR at 1/3, 1/2, 2/3 (and 4/5), in the sense of §10. Four or more fields and the full nonlinear descent were not run.

10 Item (6): what the modular ranks prove

All matrices are evaluations of candidate functionals, or of their σ_0-images, at random jet configurations reduced modulo p. Their entries are images of elements of $\mathbb{Z}[\frac{1}{D}][i]$ under $\mathbb{Z}[i] \to \mathbb{F}_p$ with $i \mapsto \sqrt{-1} \bmod p$ ($p \equiv 1 \bmod 4$; D collects the denominators of α and of the coefficients $(m \pm a)/2$, coprime to p). Let V be the true candidate span over $\mathbb{Q}(i)$, $V_p \subseteq V$ the span of the pivot candidates found mod p, $Z = \ker(\mathrm{WZ}) \subseteq V$, and $B = \sigma_0(V^{(0)})$.

Lemma 10.1 (one-sided rank bounds)**.** *Rank mod p of any such sampled matrix is at most its rank over $\mathbb{Q}(i)$, which in turn is at most the true rank of the functionals. Hence, rigorously:*

1. *the pivot candidates are linearly independent over $\mathbb{Q}(i)$, so $\dim V \geq \dim V_p = r_p$;*
2. $\dim B \geq \mathrm{rank}_p(\textit{images})$*;*
3. $\dim(Z \cap V_p) \leq \mathrm{nullity}_p(\textit{WZ rows on } V_p)$*.*

Proof. Reduction modulo a prime ideal cannot create a nonzero minor. For (3), the kernel over $\mathbb{Q}(i)$ restricted to V_p has a saturated $\mathbb{Z}[\frac{1}{D}][i]$-lattice basis whose reduction stays independent and lies in the mod-p kernel. □

Corollary 10.2 (rigorous form of every "$H^1 = 0$" in v1–v3)**.** *Assume (Span): $V = V_p$, i.e. the dependencies found among raw candidates mod p are genuine. Then $B \subseteq Z$, which holds exactly because $\sigma_0^2 = 0$ algebraically and the enumeration is complete. Together with Lemma 10.1 this gives* $\mathrm{rank}_p(\text{images}) \leq \dim B \leq \dim Z \leq \mathrm{nullity}_p$*. Every run has* $\mathrm{nullity}_p = \mathrm{rank}_p$ *(e.g. 26/26, 83/83, 120/120), so $H^1 = 0$* exactly*. Likewise $H^1 \leq 2$ holds exactly in the F diagnostics (31/29).*

The remaining probabilistic step. Only (Span) is probabilistic. A false dependency requires a nonzero $r \times r$ minor of the sampled matrix to vanish. Each entry has degree ≤ 2 (two-point) or ≤ 3 (three-point) in the configuration variables and ≤ 7 in the momenta, so the minor has degree $\leq 10r$. With $r \leq 294$, Schwartz–Zippel bounds the probability per prime by $10r/p \leq 3 \times 10^{-6}$. All runs were repeated at p_1 and p_2, and the two-point runs and the generic three-field run also at $p_3 = 10^9 + 21$, with identical outcomes. We did *not* produce a fraction-free elimination over $\mathbb{Q}(i)$, nor a determinant-minor certificate with an a-priori height bound. Either would remove (Span). **Item (6) outcome: partially upgraded**: rigorous one-sided bounds plus an isolated hypothesis, three primes; not exact.

11 Literature

Markers as in v2: * equation checked in the downloaded text; ‡ checked at abstract/bibliography level.

* [FM97] D. V. Fursaev, G. Miele, NPB 484 (1997) 697, hep-th/9605153. [S11] S. N. Solodukhin, Living Rev. Rel. 14 (2011) 8, eq. (70).

‡ [H83] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer 1983, Thm. 3.2.3 (homogeneous extension of homogeneous functions of non-exceptional degree), used in §5.

‡ [S15] S. N. Solodukhin, PRD 91 (2015) 084028; [BC24] A. Blommaert, S. Colin-Ellerin, arXiv:2405.12276; [K95] D. Kabat, NPB 453 (1995) 281.

cited input Enter companion sheets I-B-VP-1 (v3), I-B-VP-1-Ex1, I-B-VP-6 (unchanged in the oct18c intake). These are cited only. No statement of theirs is used as a premise beyond what they prove, and none is promoted.

12 Reproducibility

The code is in `/workspace/XVI-O1-CONE/code` and the logs in `logs/v3/`:

- `inner_degrees.py` and `inner_degrees_q24.py` (Thm. 5.1);
- `tadpole2.py` and `tadpole2_fast.py` (D12, D05a, D02; `tad2*_*.json`);
- `tri_count_all.py` and `tri_run_all.py` (`tri_all_*`);
- `batch_v3_p2.sh` (third prime, `logs/v2apex/*_p2.json`).

`apex_jets.PRIMES` gained $p_3 = 1000000021$; this is our own code, not frozen. The progress record is `CHECKPOINT_v3.md`. The XVI-P, XVI-Q and XVI-R tree hashes were checked before and after and are identical:

- XVI-Q `0df9a0d1...0493c4`;
- XVI-P `ec965c15...b3aaa6`;
- XVI-R `57187bc6...331edb`.

v1 and v2 are unchanged. Nothing was pushed or written to the site.

13 Ledger

Item	Status
Step 0: registry re-intake (249 ids; 0 new, 0 changed)	done; nothing new to use
Regular-region locality; FM/Kabat matches; $E = 1$ row (D14a,b,c)	as v2 (computed)
Two-point inventory, $H^0_{(2)} = 2$, $H^1_{(2)} = 0$, all α (CR)	rigorous modulo (Span); 3 primes
(1) ($\mathrm{H_{reg}}$) for bubbles, inner-non-resonant α	proved modulo (J) and standard lemmas
(1) inner-resonant set CR $\{1/2, 2/3, 4/5\}$, F $\{1/2\}$	computed, complete; obstruction stated
(2) 26 Ward-constrained pole values = 0	modulo (J) (inner-non-resonant α)
(2) bubble apex poles D01, D03, D04, D06, D11, D13; 6 exact + 2 $\phi\phi$ values	open (reduced to a Mellin residue; not evaluated)
(3) D12 apex pole	computed, exact, branch-independent

(3) D05a apex pole	computed, exact, CR and F; prefactor flagged
(3) D02 apex pole	computed, exact, CR and F
(3) map of explicit-basis bilinears to the 18-dim hh basis	not done
(4) evanescent apex coefficients	open
(5) $H^1_{(3)} = 0$ at 1/3, 1/2, 2/3, 4/5 (all jets)	computed (2 primes each; modulo (Span))
(6) rank certificates	one-sided rigorous; (Span) probabilistic; not exact
≥ 4 fields; full nonlinear descent	open
Independent review	not started (not requested)
O.1 on the cone	**NOT_CLOSED**
O.1 (flat, scoped Q11)	CLOSED (scoped), untouched
`bulk_symmetry_restored`	false

Why NOT_CLOSED. The scoped analogue of Q11 on the cone (one-loop κ^2 bulk+apex two-point sector, CR, $m = 0$, frozen gauge, the regularisation of Prop. 5.2) requires every two-point graph's apex pole to be computed, projected and shown ST-compatible. The bubble poles (item 2) are not computed, (J) is not proved, ($\mathrm{H_{reg}}$) is obstructed at $\{1/2, 2/3, 4/5\}$, the evanescent coefficients are open, and the rank certificates rest on (Span). CLOSED_PENDING_INDEPENDENT_REVIEW is therefore not claimed for any scope.

Non-claims

We claim no Einstein dynamics and no emergence. The cone and the CR extension are inputs. No separation is used as a bridge, and no direction is certified as physical. `bulk_symmetry_restored` = `false`. Q11, XVI-P, XVI-Q and XVI-R are unmodified. A missing result is never counted as zero, and partial results are reported as partial. The graviton-loop coefficients (D14a/b, D05a, D02) are new and unreviewed.